

Excel Learning, Teaching, and Reference Guide

This document is designed to work in three ways:

Learn

Use it to understand Excel concepts step by step. Each section explains what a tool is, why it matters, and where it is used at work.

Refer Back

Use it later as a reference guide to refresh your memory. You can jump straight to the stage of Excel you are working on.

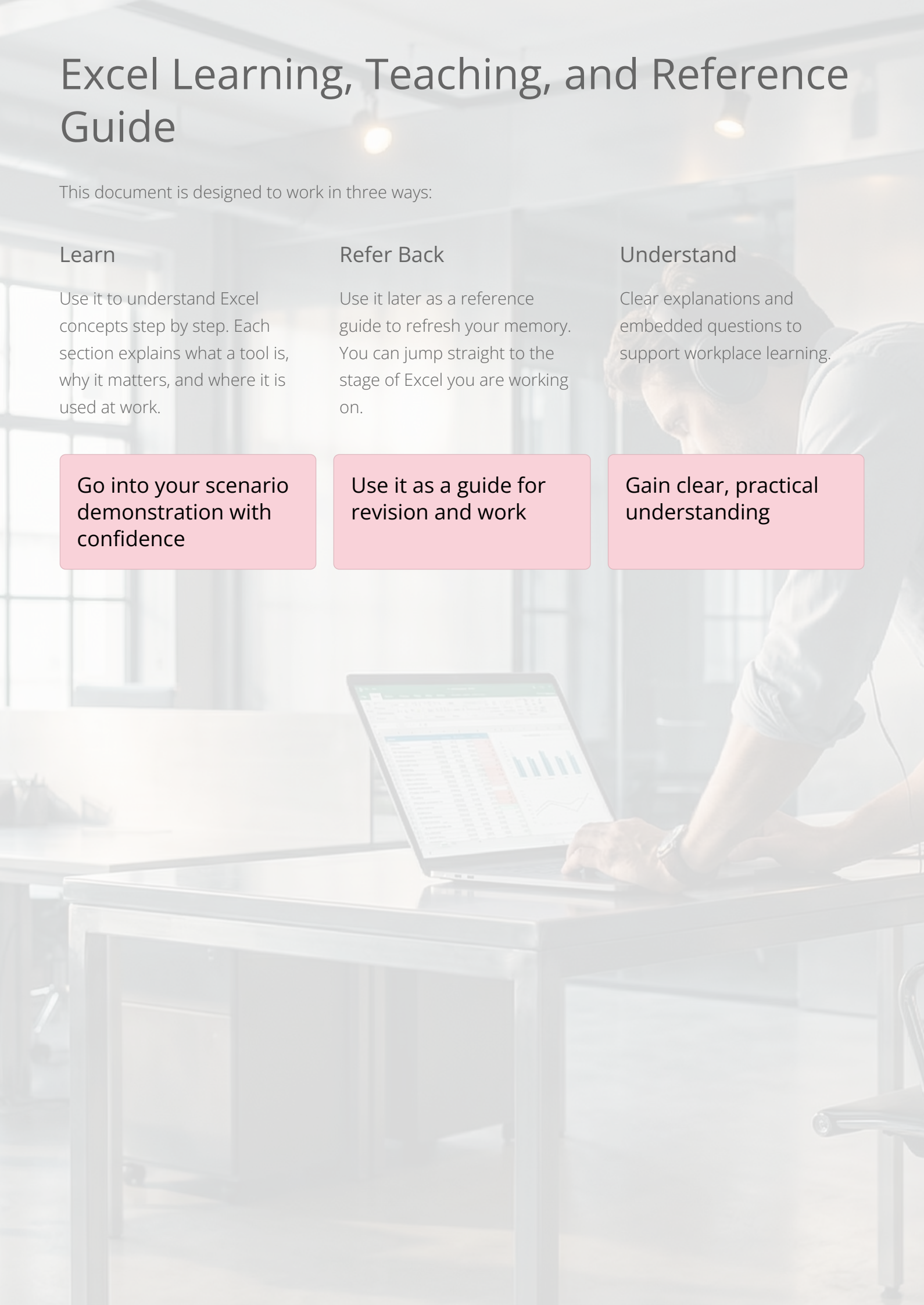
Understand

Clear explanations and embedded questions to support workplace learning.

Go into your scenario demonstration with confidence

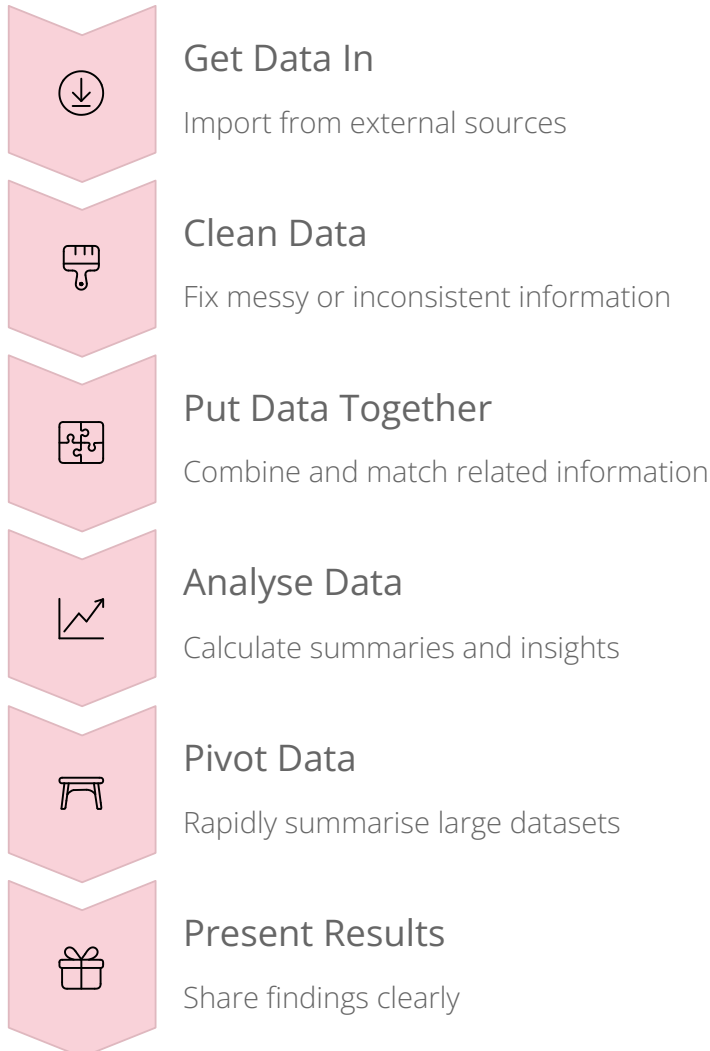
Use it as a guide for revision and work

Gain clear, practical understanding



How the Content is Organised

The guide follows the **real Excel workflow** that mirrors how Excel is used in actual jobs. This isn't about learning features in alphabetical order—it's about understanding the logical progression of working with data in a professional context.



Each stage builds on the previous one. You'll find that most Excel work follows this sequence, whether you're preparing a monthly report, analysing sales figures, or building a dashboard for your manager.

How to Get the Most Value

This guide is designed for **understanding**, not memorising formulas. Excel is a tool for solving problems, and the best way to learn is by thinking through how and why each feature works.

01

Read the explanation

Understand what the function does and why it exists. Don't skip the "Why it matters" sections—they connect features to real work.

02

Think about the questions

Each section includes reflection questions. Take a moment to think through your answer before checking the suggested response.

03

Check the suggested answers

Compare your thinking with the suggested answers. If they differ, ask yourself why.

04

Apply the ideas to your own work

Think of a situation in your job where you could use this feature. Real application is where learning sticks.

Get Data In

Before you can analyse anything, you need to get data into Excel. The **Get Data** feature is your starting point for importing information from external sources such as databases, other Excel files, CSV exports, PDFs, and web pages.

Core Feature: Get Data

What it is used for: Importing data from external sources into Excel in a structured, repeatable way.

Why it matters: Get Data reduces copying errors and makes reports repeatable. Instead of manually copying and pasting data every week or month, you set up the import once and refresh it whenever needed. This saves time and ensures consistency.

What to watch out for: Imported data almost always needs cleaning. Systems export data in formats that aren't immediately ready for analysis—dates might be text, numbers might have extra spaces, or columns might be in the wrong order.

Typical workplace example: Importing a monthly system report from your finance or HR system. Rather than copying values by hand, you use Get Data to pull the report directly into Excel.

Get Data is the foundation of modern Excel workflows. Once you understand how to import data properly, you unlock the ability to build reports that update automatically with fresh information.



Where to find it

Data tab → Get Data

This opens a menu showing all available import options.

Get Data from Workbook

Core Feature: Get Data → From Workbook

What it is used for: Pulling data from another Excel file into your current workbook. This is particularly useful when you need to reuse data that's maintained by someone else or stored in a shared location.

Why it matters: Get Data from Workbook allows you to reuse trusted data sources without duplicating information. For example, if your finance team maintains a master file of budget codes, you can link to that file rather than copying the data into your own workbook. When the master file is updated, you simply refresh your connection.

What to watch out for: Changes in the source file affect your results. If someone renames the file, moves it to a different folder, or changes the structure of the data, your connection may break. Always check file paths and communicate with data owners.

This feature is essential for maintaining data integrity across teams. Instead of everyone keeping their own copy of reference data, one authoritative source is used by all.

Syntax

Data → Get Data → From File
→ From Workbook
→ Browse to the file
→ Select the table or range
→ Load



Example

Linking to a shared finance file that contains department codes and budget holders. You use this data to validate entries in your own report.

Get Data from CSV, Text, and PDF

Core Features: Get Data → From CSV / From Text / From PDF

What they are used for: Importing system exports and locked documents. Most business systems export data as CSV (comma-separated values) or text files. Some documents, such as supplier invoices or regulatory reports, arrive as PDFs that need to be extracted into Excel for analysis.

Why it matters

Most business data arrives in these formats. If you rely on manual copying, you introduce errors and waste time. Get Data automates the process and ensures that the same steps are followed each time, making your work faster and more reliable.

CSV and text imports are especially common in HR, finance, and operations roles. PDF imports are useful when you receive locked reports from external suppliers or regulators.

What to watch out for

Dates, numbers, and layouts may import incorrectly. For example, dates might appear as text (e.g. "01/03/2024" instead of a true date value), numbers might include commas or currency symbols that prevent calculations, and data might span multiple rows or columns unexpectedly.

Always preview imported data and check for formatting issues before using it in calculations.

1

CSV Import

Data → Get Data → From File → From Text/CSV → Browse to file → Load

2

PDF Import

Data → Get Data → From File → From PDF → Browse to file → Select table → Load



Typical workplace example: Importing weekly HR absence reports exported from your system as CSV, or extracting supplier invoices received as PDF.

Refresh Data

Core Feature: Refresh

What it is used for: Updating imported data automatically. Once you've set up a Get Data connection, you don't need to repeat the import steps every time the source data changes. Instead, you click Refresh (or set Excel to refresh automatically when the file opens).

Why it matters: Refresh keeps reports current without repeating steps. Imagine you import a sales report every Monday. Without Refresh, you'd need to delete the old data, re-import the new file, and reapply any cleaning steps. With Refresh, you simply click one button and Excel updates everything automatically.

Typical workplace example: Weekly reporting where you import data every Monday morning, refresh the connection, and send the updated report to your manager.

Syntax

Data → Refresh All
(or right-click a table
→ Refresh)



What to watch out for

Broken file paths or renamed files. If the source file has been moved or renamed, Excel won't be able to find it. You'll see an error message asking you to update the connection.

Always save source files in stable, shared locations with consistent names.

1

Click to update

One button refreshes all imported data across your workbook

5min

Time saved

Average time saved per report by using Refresh instead of manual re-import

Assessment – Get Data In

Think About It

1. Why is *Get Data* safer than copy and paste?
2. When would you choose CSV instead of Workbook?
3. Why might imported data not be ready to analyse?
4. What does Refresh save you from doing manually?

Suggested Answers

1. It reduces manual errors and keeps steps repeatable. Every time you copy and paste, there's a risk of selecting the wrong range, missing rows, or introducing formatting problems. Get Data eliminates this risk.
2. CSV is used for system exports where no Excel file exists. Many systems can only export data as CSV or text, not as native Excel files.
3. Imported data may have formatting, text, or layout issues. Systems don't format data for human readability—they format it for machine processing. You'll often need to clean dates, remove extra spaces, or convert text to numbers.
4. Refresh updates reports without re-importing files. It saves you from repeating the import process every time the source data changes.

 **Key takeaway:** Get Data is the foundation of reliable, repeatable reporting. Master this stage and you'll save hours every week.

Clean Data Up

Once data is imported, it's rarely ready to use immediately. Cleaning data means fixing inconsistencies, removing errors, and standardising formats so that calculations and analysis work correctly.

Core Functions: TRIM, CLEAN

What they are used for: Fixing messy text by removing extra spaces and hidden characters. When data is copied from emails, web pages, or older systems, it often contains invisible characters or multiple spaces between words.

Why it matters: Extra characters break lookups and logic. If you're trying to match "John Smith" to a list of names, but one version has a space at the end, Excel won't recognise them as the same. TRIM and CLEAN fix this.

What to watch out for: Invisible characters you can't see. Sometimes data looks correct on screen but still won't match because of hidden line breaks or non-printing characters. CLEAN removes these.

Syntax and Examples

```
=TRIM(A1)
```

What this does: Removes all extra spaces from text in cell A1, leaving only single spaces between words.

```
=CLEAN(A1)
```

What this does: Removes non-printable characters from text in cell A1.

 **Example:** Names copied from emails often have extra spaces or line breaks. TRIM and CLEAN fix this before you use the names in a lookup or mail merge.

Fixing Capitalisation

Core Functions: UPPER, LOWER, PROPER

What they are used for: Standardising text values. Databases and systems often store names or codes in inconsistent capitalisation (e.g. "SMITH", "Smith", "smith"). These functions convert text to a consistent format.

Why it matters: Consistent text matches correctly. If you're checking whether a name appears in a list, "SMITH" and "smith" won't match unless you standardise them first.

What to watch out for: PROPER can change acronyms incorrectly. For example, "HR" becomes "Hr" and "UK" becomes "Uk". Always check for acronyms before applying PROPER to an entire column.

Typical workplace example: Customer names or staff names imported from different systems with inconsistent capitalisation.

UPPER

```
=UPPER(A1)
```

Converts text to ALL CAPITALS.

Example: "john smith" → "JOHN SMITH"

LOWER

```
=LOWER(A1)
```

Converts text to all lowercase.

Example: "JOHN SMITH" → "john smith"

PROPER

```
=PROPER(A1)
```

Converts text to Title Case.

Example: "john smith" → "John Smith"

Checking Text Length & Content

Core Functions: LEN, FIND, SEARCH

What they are used for: Validating text structure. These functions help you check whether data follows the expected pattern—for example, ensuring that all postcodes are 6–8 characters long, or that email addresses contain an "@" symbol.

Why it matters

LEN, FIND, and SEARCH identify invalid or incomplete entries. If you're importing customer data and some email addresses are missing the "@" symbol, FIND can flag these for correction before you send communications.

These functions are essential for data quality checks.

What to watch out for

FIND is case-sensitive; SEARCH is not. If you're looking for "HR" in a text string, FIND will only match "HR" exactly. SEARCH will match "hr", "Hr", or "HR".

Choose the function that matches your needs.

LEN

```
=LEN(A1)
```

Returns the number of characters in cell A1.

Example: "SW1A 1AA" returns 8.

FIND

```
=FIND("@", A1)
```

Returns the position of "@" in cell A1 (case-sensitive).

Returns an error if not found.

SEARCH

```
=SEARCH("@", A1)
```

Returns the position of "@" in cell A1 (not case-sensitive).



Typical workplace example: Email address checks to ensure all addresses are valid before sending a bulk email campaign.

Replacing or Converting Data

Core Functions: SUBSTITUTE, REPLACE, VALUE, TEXT

What they are used for: Correcting and converting data. Sometimes you need to swap one piece of text for another (e.g. replacing "Ltd" with "Limited"), or convert text that looks like a number into an actual number that Excel can calculate with.

Why it matters

These functions enable calculations and analysis. If sales figures are imported as text (e.g. "1,234" with a comma), Excel can't add them up. VALUE converts text to numbers. TEXT does the opposite, converting numbers to formatted text when needed (e.g. for concatenating values with labels).

What to watch out for

Formatting does not change the underlying value. If a number is stored as text, making it *look* like a number with formatting won't fix it—you must convert it with VALUE.

Similarly, TEXT creates text, not a number you can calculate with.

SUBSTITUTE

```
=SUBSTITUTE(A1, "Ltd", "Limited")
```

Replaces "Ltd" with "Limited" in cell A1.

REPLACE

```
=REPLACE(A1, 1, 3, "ABC")
```

Replaces the first 3 characters in cell A1 with "ABC".

VALUE

```
=VALUE(A1)
```

Converts text that looks like a number into an actual number.

TEXT

```
=TEXT(A1, "£#,##0")
```

Converts the number in A1 to formatted text (e.g. "£1,234").

Typical workplace example: Imported sales figures where values are text with commas. VALUE converts them to numbers for SUM calculations.

Removing Duplicates & Controlling Input

Core Features: UNIQUE, SORT, FILTER, Data Validation

What they are used for: Simplifying and protecting data. UNIQUE removes duplicate rows so you have one entry per customer, product, or transaction. SORT and FILTER help you organise and view specific subsets of data. Data Validation prevents users from entering invalid information (e.g. restricting a cell to only accept dates, or only values from a specific list).

Why it matters

These features prevent double-counting and bad input. If your dataset contains duplicate customer records, your analysis will be wrong. Removing duplicates ensures accurate counts and summaries.

Data Validation stops errors before they happen by restricting what users can type into cells.

What to watch out for

Removing duplicates permanently deletes rows. Always work on a copy of your data, or use UNIQUE (which doesn't delete anything—it just returns unique values in a new location).

Data Validation only works if users don't copy and paste data from elsewhere, which bypasses validation rules.

UNIQUE

```
=UNIQUE(A1:A100)
```

Returns a list of unique values from range A1:A100.

SORT

```
=SORT(A1:A100)
```

Returns the range A1:A100 sorted alphabetically.

FILTER

```
=FILTER(A1:C100, B1:B100="Active")
```

Returns rows from A1:C100 where column B equals "Active".

Data Validation

Data tab → Data Validation → Allow: List → Source: range or list of values



Typical workplace example: Ensuring one row per customer in a mailing list, or restricting a "Department" column to only accept values from a predefined list.

Assessment – Clean Data Up

Think About It

1. Why can extra spaces cause lookups to fail?
2. When is PROPER not a good choice?
3. What's the difference between formatting and converting a number?
4. Why should duplicates be removed carefully?

Suggested Answers

1. Extra spaces stop exact matches from working. Excel treats "John Smith" and "John Smith " (with a trailing space) as different values, even though they look the same.
2. PROPER can damage acronyms like UK or HR. It will convert them to "Uk" and "Hr", which looks unprofessional and may break logic that relies on uppercase codes.
3. Formatting only changes appearance; conversion changes the underlying value. A number stored as text won't work in calculations until you convert it with VALUE.
4. Removing duplicates deletes data permanently. Always work on a copy, or use UNIQUE instead, which doesn't alter the original data.

 **Key takeaway:** Clean data is the foundation of accurate analysis. Invest time in cleaning properly and you'll avoid errors downstream.

Put Data Together

Once your data is clean, you often need to combine information from different sources or extract specific parts of text to create useful fields for analysis. This stage is about joining, matching, and restructuring data.

Core Functions: LEFT, RIGHT, MID

What they are used for: Pulling structured data from text. For example, if you have a product code like "AB-1234-XY" and you need to extract the "AB" (category), "1234" (item number), or "XY" (variant), you use LEFT, RIGHT, or MID to split the code into separate fields.

Why it matters: Extracting parts of text enables grouping and matching. Instead of analysing individual product codes, you can group by category or variant. This is essential for summarising and reporting.

What to watch out for

Inconsistent formats. If some product codes are "AB-1234-XY" and others are "AB1234XY" (no hyphens), your formulas will return incorrect results. Always check that your data follows a consistent pattern before extracting text.



Typical workplace example:

Extracting postcode districts (e.g. "SW1A" from "SW1A 1AA") or department codes from employee ID numbers.

LEFT

```
=LEFT(A1, 2)
```

Returns the first 2 characters from cell A1.

Example: "AB-1234" → "AB"

RIGHT

```
=RIGHT(A1, 4)
```

Returns the last 4 characters from cell A1.

Example: "AB-1234" → "1234"

MID

```
=MID(A1, 4, 4)
```

Returns 4 characters starting at position 4 in cell A1.

Example: "AB-1234-XY" → "1234"

Joining Text

Core Functions: CONCAT, &, TEXTJOIN

What they are used for: Combining text fields. You might need to join a first name and surname into a full name, or combine address lines into a single field for mail merge. These functions bring separate pieces of text together into one cell.

Why it matters

Joining text creates readable identifiers. Instead of working with separate "First Name" and "Last Name" columns, you create a single "Full Name" field that's easier to read and use in reports.

TEXTJOIN is particularly useful because it allows you to add separators (like spaces or commas) and ignore empty cells automatically.

What to watch out for

Missing spaces or separators. If you join "John" and "Smith" without a space, you get "JohnSmith". Always include separators where needed.

The & operator is simple but requires you to type separators manually. TEXTJOIN handles this automatically.



CONCAT

```
=CONCAT(A1, " ", B1)
```

Joins cell A1, a space, and cell B1.

Example: "John" + " " + "Smith" → "John Smith"



& Operator

```
=A1 & " " & B1
```

Same as CONCAT but using the & symbol.

Example: "John" & " " & "Smith" → "John Smith"



TEXTJOIN

```
=TEXTJOIN(", ", TRUE, A1:A5)
```

Joins cells A1 to A5 with commas, ignoring blanks.

Example: "Red", "Blue", "", "Green" → "Red, Blue, Green"



Typical workplace example: Creating full names from first and last name columns, or joining multiple address lines into a single field for printing labels.

Looking Up Related Data

Core Functions: XLOOKUP, VLOOKUP

What they are used for: Pulling data from other tables. Lookup functions let you match a value in one table (e.g. an employee ID) and retrieve related information from another table (e.g. their department or manager). This avoids duplicate data entry and keeps information consistent.

Why it matters

Lookups avoid duplicate data entry. Instead of manually typing department names next to every employee ID, you maintain one master list and use a lookup to pull the correct department automatically.

XLOOKUP is the modern, flexible version of VLOOKUP. It's easier to use and more powerful.

What to watch out for

Missing or duplicate lookup values. If an employee ID doesn't exist in the master list, the lookup returns an error. If the same ID appears twice in the master list, Excel returns the first match, which may not be the one you want.

Always check your lookup table for completeness and uniqueness.



XLOOKUP

```
=XLOOKUP(A2, EmployeeIDs,  
Departments)
```

What this does: Looks for the value in A2 within the EmployeeIDs range, and returns the corresponding value from the Departments range.

Example: Employee ID "E123" → "Finance"



VLOOKUP

```
=VLOOKUP(A2, Table, 3, FALSE)
```

What this does: Looks for the value in A2 in the first column of Table, and returns the value from the 3rd column. FALSE means exact match.

Example: Employee ID "E123" → "Finance"



Typical workplace example: Looking up an employee ID to retrieve their department, manager, or cost centre from a master staff list.

Advanced Lookups

Core Functions: INDEX, MATCH

What they are used for: Flexible lookup logic. INDEX and MATCH work together to create lookups that don't depend on column position. This is useful when your table structure changes (e.g. columns are added or rearranged), or when you need to look left (VLOOKUP can only look right).

Why it matters

INDEX and MATCH work when table layouts change. VLOOKUP breaks if you insert a column before your lookup column, or if you need to return a value that's to the left of your lookup column. INDEX and MATCH don't have these limitations.

This combination is more robust for reports that need to be maintained over time.

What to watch out for

Incorrect match logic. MATCH has three modes: exact match (0), less than or equal to (-1), and greater than or equal to (1). Using the wrong mode will return incorrect results.

For most lookups, use exact match (0).



MATCH

```
=MATCH(A2,  
EmployeeIDs, 0)
```

Finds the position of A2 within EmployeeIDs (exact match).

Returns a number (e.g. 5 if the ID is in the 5th row).



INDEX

```
=INDEX(Departments, 5)
```

Returns the value in the 5th position of the Departments range.



Combined

```
=INDEX(Departments,  
MATCH(A2, EmployeeIDs,  
0))
```

Looks up A2 in EmployeeIDs, finds its position, and returns the corresponding value from Departments.



Typical workplace example: Dynamic reports where column order may change, or lookups where the return value is to the left of the lookup value.

Assessment – Put Data Together

Think About It

1. When would you extract text instead of rewriting it?
2. Why is XLOOKUP preferred over VLOOKUP?
3. What problem do INDEX and MATCH solve together?

Suggested Answers

1. When text follows a consistent pattern. If all product codes are structured as "XX-1234-YY", you can extract the parts automatically instead of manually typing them. This is faster and eliminates errors.
2. XLOOKUP is more flexible and easier to maintain. It doesn't break when you insert columns, it can look left, and it has built-in error handling. VLOOKUP requires you to count column positions, which is fragile and error-prone.
3. They allow lookups that don't depend on column position. INDEX and MATCH let you look up values even if columns are rearranged or if the return value is to the left of the lookup value. This makes reports more robust and maintainable.

 **Key takeaway:** Combining and matching data is essential for analysis. Master lookups and text functions, and you'll be able to work with data from multiple sources efficiently.

Analyse Data

Now that your data is clean and combined, you can start analysing it. This stage is about making decisions, calculating summaries, and generating insights that drive action.

Core Operators & Functions: =, >, <, >=, <=, <>, IF, IFS, AND, OR, NOT

What they are used for: Decision-making logic.

Comparison operators (=, >, <) test whether values meet certain conditions. IF, IFS, AND, OR, and NOT turn these tests into automated decisions, applying different outcomes based on rules.

Why it matters: These functions automate judgement rules. Instead of manually checking whether each student passed or failed, you write a formula that applies the rule to every row. This is faster, more consistent, and easier to audit.

What to watch out for

Over-complex formulas. Deeply nested IF statements become difficult to read and maintain. If you find yourself nesting more than three IFs, consider using IFS instead, or break the logic into separate helper columns.



Typical workplace example:

Grading student results (Pass, Merit, Distinction) based on score ranges, or categorising invoices as "Overdue", "Due Soon", or "Paid".

IF

```
=IF(A2>=70, "Pass", "Fail")
```

If A2 is 70 or more, return "Pass". Otherwise, return "Fail".

IFS

```
=IFS(A2>=70, "Distinction", A2>=50, "Merit", A2>=40, "Pass", TRUE, "Fail")
```

Tests multiple conditions in order. Returns the result for the first condition that's true.

AND

```
=IF(AND(A2>=50, B2="Submitted"), "Approved", "Pending")
```

Both conditions must be true. Useful for multi-criteria decisions.

Basic Summaries

Core Functions: SUM, AVERAGE, COUNT, COUNTA, MIN, MAX, ROUND

What they are used for: Numeric summaries. These are the most fundamental analysis functions. SUM adds values, AVERAGE calculates means, COUNT counts numbers, MIN and MAX find extremes, and ROUND controls decimal places for presentation.

Why it matters

Basic summaries are the foundation of reporting. Nearly every analysis starts with totals, averages, or counts. Understanding these functions is essential before moving to more advanced techniques.

ROUND is often overlooked but critical for making reports readable and consistent.

What to watch out for

Incorrect ranges. If you select the wrong range in a SUM or AVERAGE, your totals will be wrong and you may not notice. Always double-check that your range includes all relevant rows and excludes header rows.

COUNT only counts cells with numbers. COUNTA counts any non-empty cell.

SUM <code>=SUM(A1:A10)</code> Adds all numbers in the range A1 to A10.	AVERAGE <code>=AVERAGE(A1:A10)</code> Calculates the mean of values in A1 to A10.
COUNT <code>=COUNT(A1:A10)</code> Counts how many cells in A1 to A10 contain numbers.	COUNTA <code>=COUNTA(A1:A10)</code> Counts non-empty cells (numbers or text).
MIN / MAX <code>=MIN(A1:A10)</code> <code>=MAX(A1:A10)</code> Returns the smallest or largest value in the range.	ROUND <code>=ROUND(A1, 2)</code> Rounds A1 to 2 decimal places.

Conditional Analysis

Core Functions: SUMIF, COUNTIF, AVERAGEIF, SUMIFS, COUNTIFS, AVERAGEIFS

What they are used for: Targeted analysis. These functions calculate summaries only for rows that meet specific criteria. For example, total sales for the "North" region, or average score for students who attended more than 80% of sessions.

Why it matters

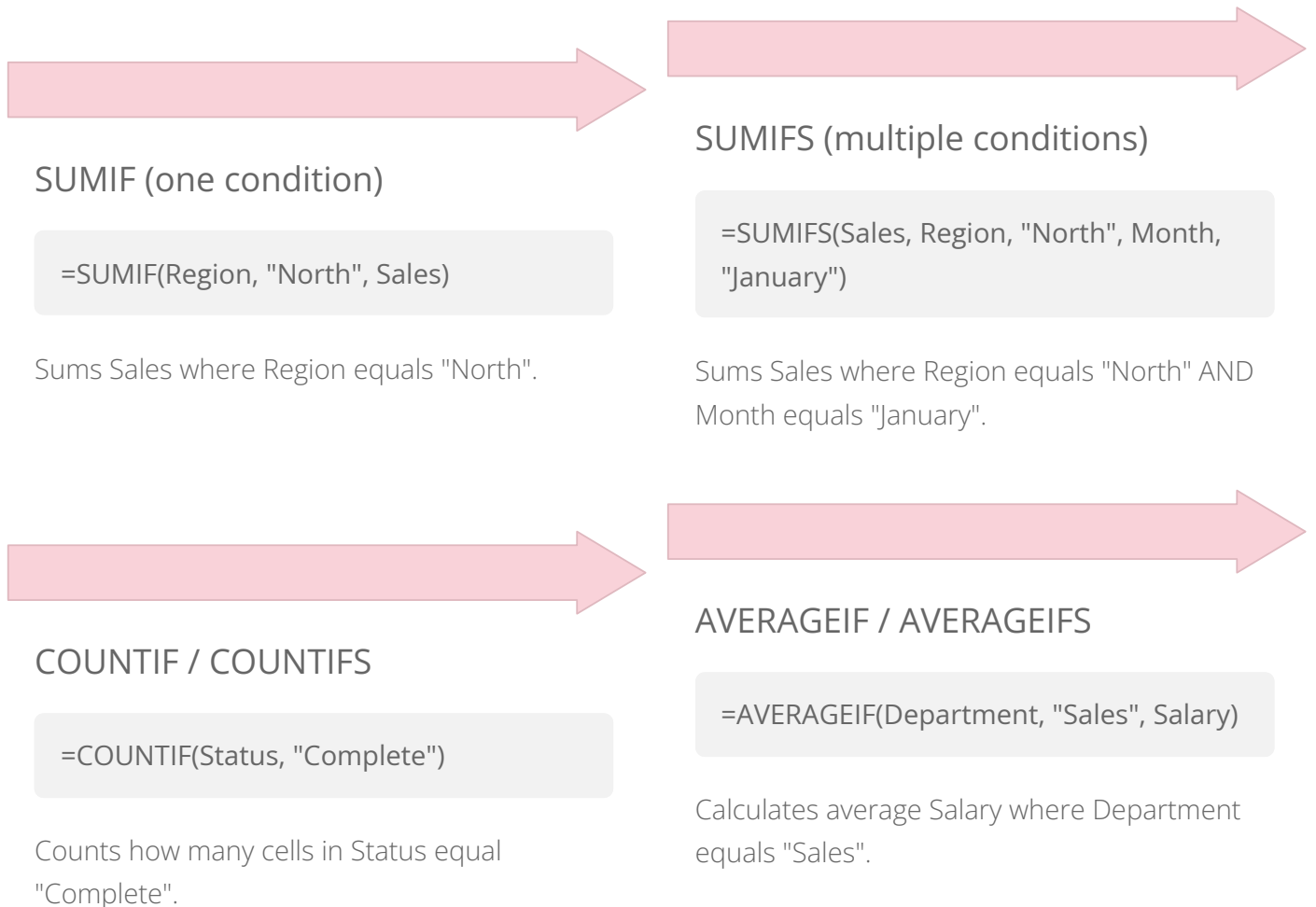
Conditional functions enable meaningful insight. Instead of calculating one grand total, you can break down totals by category, region, product, or any other dimension. This is where analysis becomes useful for decision-making.

The "S" versions (SUMIFS, COUNTIFS, AVERAGEIFS) allow multiple criteria, making them extremely powerful.

What to watch out for

Criteria ranges must align. If your data is in rows 2 to 100, your criteria range must also be rows 2 to 100. Misaligned ranges produce incorrect results.

Criteria are case-insensitive by default. "north", "North", and "NORTH" are treated as the same.



 **Typical workplace example:** Sales by region and month, or average attendance by course and term.

Assessment – Analyse Data

Think About It

1. When would you use IFS instead of nested IFs?
2. Why are conditional functions better than filtering manually?
3. What happens if criteria ranges don't match?

Suggested Answers

1. IFS is clearer and easier to maintain. When you have three or more conditions, nested IF statements become hard to read and debug. IFS lists all conditions clearly in sequence, making the logic easier to understand and modify.
2. Conditional functions update automatically and reduce errors. Filtering manually is time-consuming and doesn't update when data changes. SUMIF and COUNTIF recalculate automatically whenever the source data is updated, ensuring your summaries are always current.
3. Excel returns incorrect results or errors. If your data range and criteria range don't align (e.g. data is rows 2–100 but criteria is rows 1–99), Excel will calculate using mismatched rows, producing wrong totals. Always check that ranges are identical.



Key takeaway: Conditional analysis is where Excel becomes genuinely powerful. Learn these functions well and you'll be able to answer almost any business question.

Pivot Data

Pivot Tables are one of Excel's most powerful features. They let you summarise, group, and analyse large datasets in seconds, without writing formulas.

Core Feature: Pivot Table

What it is used for: Rapid summarisation of large datasets. Pivot Tables let you drag and drop fields to create summaries by any dimension—region, product, month, department, or any combination. You can change the summary structure instantly by moving fields around.

Why it matters: Pivot Tables save significant time. What would take dozens of formulas and manual grouping can be done in a Pivot Table in under a minute. They're essential for exploratory analysis and recurring reports.

What to watch out for

Dirty source data. Pivot Tables are powerful but they don't clean your data for you. If your source data contains extra spaces, inconsistent spellings, or blank rows, your Pivot Table will reflect those errors.

Always clean your data before creating a Pivot Table.



Typical workplace example:

Sales by region, product sales by month, or staff headcount by department and grade.

01

Select your data

Click anywhere in your data table. Excel will detect the range automatically.

03

Drag fields to areas

Drag dimensions (Region, Product) to Rows or Columns. Drag numeric fields (Sales) to Values.

02

Insert Pivot Table

Insert tab → Pivot Table → OK to create on a new sheet.

04

Customise

Change the summary type (Sum, Average, Count), add filters, or change field order to explore your data.

Pivot Charts & Refreshing

Core Features: Pivot Chart, Refresh Pivot

What they are used for: Visualising and updating summaries. Once you've created a Pivot Table, you can add a Pivot Chart to visualise the data. Pivot Charts update dynamically as you change the Pivot Table structure. Refresh updates the Pivot Table when the source data changes.

Why it matters

Pivot Charts keep insights current. Instead of creating separate charts from formulas, Pivot Charts automatically reflect your Pivot Table summaries. When you change the Pivot Table (e.g. switch from Region to Product), the chart updates instantly.

Refreshing ensures your Pivot Table shows the latest data. Without refreshing, your Pivot Table becomes out of date.

What to watch out for

Forgetting to refresh. Pivot Tables don't update automatically when source data changes. You must manually refresh (or set Excel to refresh on file open).

Always check the refresh date before relying on a Pivot Table summary.



Create Pivot Chart

Click anywhere in the Pivot Table → PivotTable Analyse tab → Pivot Chart → Choose chart type → OK



Refresh Pivot Table

Right-click the Pivot Table → Refresh
Or: PivotTable Analyse tab → Refresh



Typical workplace example: Monthly sales trends visualised in a bar chart, automatically updating when you switch from Region to Product view.

Extracting Pivot Values

Core Function: GETPIVOTDATA

What it is used for: Pulling stable values from Pivot Tables. GETPIVOTDATA retrieves a specific value from a Pivot Table based on field names and criteria. Unlike a regular cell reference, GETPIVOTDATA continues to work even if the Pivot Table layout changes.

Why it matters

GETPIVOTDATA creates reliable dashboards. If you build a KPI dashboard that pulls values from a Pivot Table, regular cell references (like =B5) will break if someone changes the Pivot Table structure. GETPIVOTDATA retrieves values by name, so it's much more robust.

What to watch out for

Behaviour changes if Pivot Table structure changes significantly. GETPIVOTDATA is stable for layout changes (adding or removing rows), but if you delete or rename a field, the formula will return an error.

Field names must match exactly, including spaces and capitalisation.

Syntax

```
=GETPIVOTDATA("Sales", A3, "Region", "North")
```

What this does: Retrieves the "Sales" value from the Pivot Table starting at cell A3, where "Region" equals "North".

Auto-generation

Type **=** in a cell, then click a value in the Pivot Table. Excel automatically generates a GETPIVOTDATA formula for you.

 **Typical workplace example:** A KPI dashboard showing "Total Sales – North Region" pulled from a Pivot Table. The formula continues to work even if rows are added or removed from the Pivot Table.

Assessment – Pivot Data

Think About It

1. Why are Pivot Tables faster than formulas?
2. Why must Pivot Tables be refreshed?
3. When is GETPIVOTDATA useful?

Suggested Answers

1. They summarise data automatically. Pivot Tables use Excel's built-in summarisation engine, which is optimised for speed. Creating the same summaries with formulas would require dozens of SUMIFS or array formulas, which are slower to build and slower to calculate.
2. They don't update unless refreshed. Pivot Tables take a snapshot of the source data when created. If the source data changes (new rows added, values updated), the Pivot Table continues to show the old snapshot until you manually refresh it.
3. It pulls stable values into reports and dashboards. If you're building a dashboard that references Pivot Table values, GETPIVOTDATA ensures the references don't break when the Pivot Table structure changes. It's more robust than cell references like =B5.

 **Key takeaway:** Pivot Tables are essential for rapid analysis. Learn to use them confidently and you'll save hours every week.

Feedback & Present Results

The final stage is presenting your analysis clearly and professionally. This includes handling errors gracefully, using visual formatting to highlight key information, and ensuring formulas behave correctly when data is filtered or copied.

Core Function: IFERROR

What it is used for: Hiding technical errors. When formulas encounter errors (e.g. a lookup value doesn't exist, or division by zero), Excel displays error codes like #N/A or #DIV/0!. IFERROR replaces these error codes with custom messages, making reports look professional.

Why it matters: IFERROR improves professional appearance. Instead of showing "#N/A" to your manager or client, you can display "Not available" or leave the cell blank.

What to watch out for

Masking real problems. IFERROR is helpful for expected errors (e.g. a new employee ID that hasn't been added to the lookup table yet), but it can hide genuine formula mistakes. If you use IFERROR everywhere, you might not notice when a formula is broken.

Use IFERROR selectively, and always test your formulas without it first.

Syntax

```
=IFERROR(XLOOKUP(A2, IDs, Names), "Not  
found")
```

If the XLOOKUP returns an error, display "Not found" instead.

Blank instead of message

```
=IFERROR(XLOOKUP(A2, IDs, Names), "")
```

If the XLOOKUP returns an error, display nothing (empty cell).

📌 **Typical workplace example:** A report showing employee names. If an employee ID is new and not yet in the lookup table, display "Pending" instead of #N/A.

Visual Feedback

Core Feature: Conditional Formatting

What it is used for: Highlighting key values automatically. Conditional Formatting applies colours, icons, or data bars to cells based on their values. For example, you can highlight overdue invoices in red, or show sales performance with colour scales (green for high, red for low).

Why it matters

Conditional Formatting enables faster understanding. Instead of scanning a list of numbers to find the highest or lowest, colour formatting makes patterns immediately visible. This is essential for dashboards and reports that need to be understood quickly.

What to watch out for

Overuse reduces clarity. If every column has different colours and icons, the formatting becomes distracting rather than helpful. Use Conditional Formatting strategically to highlight only the most important information.

Keep it simple and consistent.



Colour Scales

Applies a gradient of colours (e.g. green to red) based on cell values. Useful for showing performance at a glance.

Home tab → Conditional Formatting → Color Scales



Data Bars

Displays horizontal bars inside cells, proportional to their values. Makes comparisons easy without a separate chart.

Home tab → Conditional Formatting → Data Bars



Highlight Cells Rules

Highlights cells that meet specific criteria (e.g. greater than 100, or dates in the last 7 days).

Home tab → Conditional Formatting → Highlight Cells Rules

 **Typical workplace example:** Highlighting overdue invoices in red, or showing sales performance with colour scales where high sales are green and low sales are red.

Responsive Totals & Formula Control

Core Functions & Concepts: SUBTOTAL, Absolute vs Relative References (\$)

What they are used for: Accurate filtered totals and safe copying. SUBTOTAL calculates sums, averages, or counts that respond correctly to filters—hidden rows are excluded automatically. Absolute references (\$) lock specific cells when copying formulas, preventing them from shifting incorrectly.

Why it matters

SUBTOTAL prevents reporting errors. If you use SUM on filtered data, it includes hidden rows in the total, which is almost never what you want. SUBTOTAL only sums visible rows, ensuring accuracy.

Absolute references (\$) prevent formulas changing incorrectly when copied. This is essential for maintaining formula integrity across large reports.

What to watch out for

Using SUM instead of SUBTOTAL on filtered data. This is a common mistake that produces incorrect totals. Always use SUBTOTAL if your data might be filtered.

Forgetting \$ symbols when copying formulas that reference fixed cells (e.g. tax rates or exchange rates stored in one cell).

SUBTOTAL

```
=SUBTOTAL(9, A2:A100)
```

What this does: Sums values in A2:A100, but excludes hidden rows. The number 9 specifies SUM. Use 1 for AVERAGE, 2 for COUNT, etc.

Relative Reference

```
=A1*B1
```

When copied down, this becomes =A2*B2, =A3*B3, etc. The reference shifts.

Absolute Reference

```
=A1*$B$1
```

When copied down, this becomes =A2*\$B\$1, =A3*\$B\$1, etc. The \$B\$1 reference stays fixed.

Mixed Reference

```
=A1*$B1
```

The column is fixed (\$B) but the row shifts (1 → 2 → 3). Useful for structured table lookups.

📌 **Typical workplace example:** Using SUBTOTAL for filtered sales reports, or using \$C\$5 to reference a fixed VAT rate when calculating tax on multiple rows.

Assessment – Feedback & Presentation

Think About It

1. Why is IFERROR helpful but risky?
2. When should SUBTOTAL be used instead of SUM?
3. Why are absolute references important?

Suggested Answers

1. It improves appearance but can hide real issues. IFERROR makes reports look professional by replacing error codes with friendly messages. However, if you use it everywhere, you might not notice when a formula is genuinely broken. Always test formulas without IFERROR first to ensure they work correctly.
2. When totals must respond to filters. If your data might be filtered (e.g. showing only "Active" customers or "Q1" sales), SUBTOTAL ensures the total only includes visible rows. SUM includes all rows, even hidden ones, which produces incorrect totals on filtered data.
3. They prevent formulas changing incorrectly when copied. If a formula references a fixed value (e.g. a tax rate in cell C5), using \$C\$5 ensures the reference stays locked when you copy the formula down or across. Without the \$ symbols, the reference would shift to C6, C7, etc., which breaks the calculation.

 **Key takeaway:** Professional presentation is about clarity and reliability. Use error handling, visual formatting, and correct formula references to make your work polished and trustworthy.

How to Use This Document

This guide is designed to support learning, teaching, and ongoing reference. Return to it whenever you need to refresh your understanding or support others in developing their Excel skills.



Learn Excel Concepts Properly

Read each section in order to build understanding progressively. Follow the real Excel workflow: get data in, clean it, put it together, analyse it, pivot it, and present results.



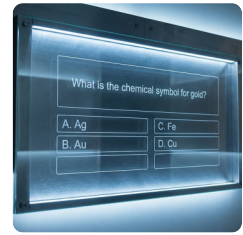
Teach Excel with Confidence

Use the clear explanations, practical examples, and embedded questions to support training sessions. The "Think about it" sections work well as discussion prompts.



Return as a Trusted Reference

Jump directly to the section you need when working on a specific task. Each section is self-contained with syntax, examples, and watch-outs.



Reinforce Understanding with Assessment

Use the assessment cards to check comprehension. Compare your thinking with the suggested answers to deepen understanding.

Key Principles

- Understanding beats memorisation
- Context matters more than syntax
- Clean data is the foundation of everything
- Formulas should be readable and maintainable

Next Steps

Apply these concepts to your own work. Start with simple tasks and build confidence. When you encounter a problem, return to the relevant section to refresh your memory.

Excel proficiency develops through practice and repetition. Keep learning, keep building, and don't be afraid to experiment.